

PRODUCT INTRODUCTION

Spectrum MK Flip-Chip COB

Indoor fine-pitch video wall cabinets for control rooms, boardrooms, retail, hospitality, and broadcast. Front service. NovaStar-ready. Stocked and supported from Azusa, California.

P1.25

P0.93

P0.78 custom

600 × 337.5 × 39.5 mm

≤ 5 kg

Front service

1,000 nits

P0.93 calibrated max

10,000:1

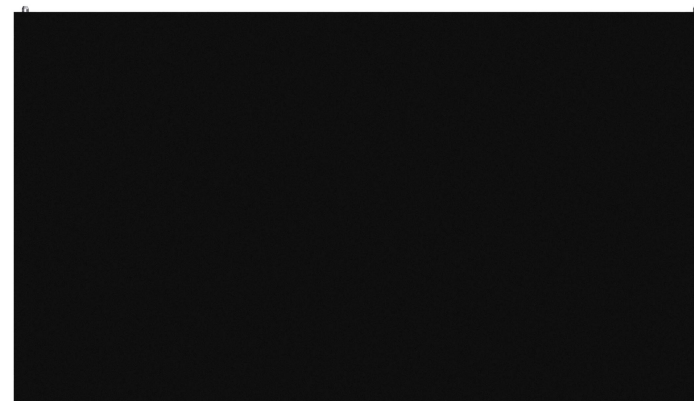
Typical contrast

≥ 160°

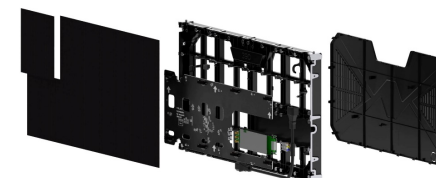
H / V viewing angle

≥ 90% DCI-P3

Color gamut



Cabinet face · 16:9 fine-pitch COB



Exploded view · modules, chassis, rear cover

CABINET ARCHITECTURE

Two electronics layouts. One 16:9 cabinet.

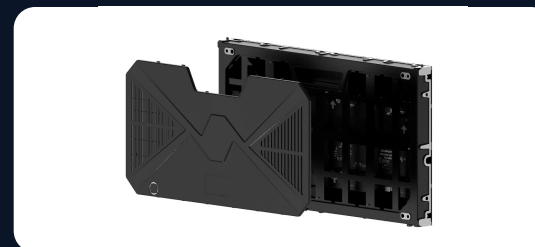
600 × 337.5 × 39.5 mm · eight 150 × 168.75 mm modules · front service · combo or separated electronics

**Module face**

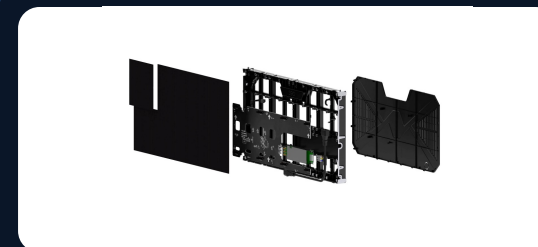
150 × 168.75 mm lamp board

**Cabinet rear**

Die-cast chassis, handle, vents

**Service access**

Rear cover lifts for PSU / cards

**Exploded stack**

Modules + frame + cover

OPTION A

HUB + receiving card combo

HUB board and receiving card share one assembly. Fewer parts in the cabinet, faster swap on a live wall. Typical for new fixed installs when the spare plan is “swap the combo unit.” Receiving cards stay inside the wall price — Spectrum sells cards only as replacements.

Fewer field parts

OPTION B

Separated HUB / card / PSU

HUB, receiving card, and power supply are independent. Service one component without pulling the others. Use this when the integrator wants a spare kit of cards and supplies, or when a control-room wall must stay up while a single part is replaced.

Independent service parts

PICTURE PERFORMANCE

High brightness. Deeper black.

Calibrated output stays readable in bright rooms without washing the blacks. Contrast holds above 10,000:1 so maps, camera feeds, and branded content stay punchy on a fine-pitch wall.

BRIGHTNESS

Up to 1,000 nits

P0.93 calibrated maximum 1,000 nits. P1.25 calibrated maximum 800 nits. Adjustable 600–1,000 nits (P0.93 / P0.78) and 600–800 nits (P1.25) so the same wall can run a dark control room or a lit lobby.

Typical market panel

≈ 600 nits picture

Spectrum MK P1.25

800 nits calibrated max

Spectrum MK P0.93

1,000 nits calibrated max

CONTRAST

Greater than 10,000:1

Flip-chip COB plus a dark surface treatment keep the panel black when the wall is off or at low APL. Color stays saturated instead of milky — useful on news, sports, and dark UI.

≥160°

Horizontal / vertical viewing

5,000–10,000 K

Adjustable color temperature

≥90% DCI-P3

Wide color gamut

3,840 Hz

Refresh, all listed pitches

GRAY SCALE AND SURFACE

Low brightness without lost gray. Even ink.

LOW-BRIGHT, HIGH GRAY

22-bit+ processing on the card

A custom receiving card runs a 22-bit+ algorithm so the wall does not posterize when you dim it for a dark room. Gradients stay even. Cabinet-to-cabinet match holds at the brightness operators actually use — not only at the demo setting.

Typical panel

Gray steps drop out as brightness falls. Banding on skies, UI, and night scenes.

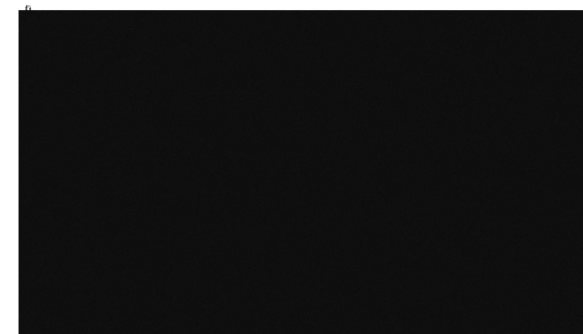
Spectrum MK

Custom card + 22-bit+ keeps gray steps at low APL. Smoother ramps, tighter match.

INK / BLACK CONSISTENCY

Darker, more even off-state

A dedicated surface process tightens module-to-module ink color. The wall looks blacker and more uniform when idle — the first thing a client notices on a fine-pitch install.



Uniform COB face — the surface that holds the black

EFFICIENCY AND PROCESSING

Lower watts. A picture engine on the card.

Flip-chip COB runs cooler than a typical SMD fine-pitch wall at the same brightness. Picture processing lives on the custom receiving card.

POWER · P1.25 REFERENCE

Less heat in the room

MARKET TYPICAL

~300W/m² peak**~110**W/m² average

SPECTRUM MK

<250W/m² peak**<85**W/m² average

Full family (table, per m²): P1.25 250 / 85 P0.93 275 / 90 P0.78 300 / 100 (max / average). Cabinet is 0.2025 m².

Lower average load means less HVAC in a sealed control room and quieter racks. Pair with NovaStar COEX (MX2000 Pro class) for the processor side — Spectrum quotes the wall and the processor together.

PICTURE QUALITY ENGINE

On the receiving card — not a PC filter

Spectrum MK uses a custom receiving card with a picture-quality engine. Market-typical cabinets at this pitch often ship a generic card with no engine. Detail, color, and low-bright behavior are processed at the cabinet.

22-bit+ gray

Keeps ramps clean when the wall is dimmed.

Picture engine

Cabinet-side processing, not a software overlay.

3,840 Hz refresh

Camera-friendly for broadcast and capture.

NovaStar path

Receiving cards included in the wall. Spares only as replacements.

SPECIFICATIONS

Spectrum MK Flip-Chip COB

Parameter	P1.25	P0.93	P0.78 custom
Package	Flip-chip COB	Flip-chip COB	Flip-chip COB
Pixel pitch (mm)	1.25	0.9375	0.78125
Module resolution (W×H)	120 × 135	160 × 180	192 × 216
Module size (mm)	150 × 168.75	150 × 168.75	150 × 168.75
Cabinet resolution (W×H)	480 × 270	640 × 360	768 × 432
Cabinet size (mm)	600 × 337.5 × 39.5	600 × 337.5 × 39.5	600 × 337.5 × 39.5
Cabinet weight	≤ 5 kg (single PSU, single system)	≤ 5 kg	≤ 5 kg
Service	Front service	Front service	Front service
Brightness (nits)	600–800 adjustable	600–1,000 adjustable	600–1,000 adjustable
Calibrated max (nits)	800	1,000	—
Color temperature (K)	5,000–10,000	5,000–10,000	5,000–10,000
Viewing angle H / V	≥ 160° / ≥ 160°	≥ 160° / ≥ 160°	≥ 160° / ≥ 160°
Gray scale	22-bit+ algorithm	22-bit+ algorithm	22-bit+ algorithm
Refresh (Hz)	3,840	3,840	3,840
Color gamut (DCI-P3)	≥ 90%	≥ 90%	≥ 90%
Max / avg power (W/m²)	250 / 85	275 / 90	300 / 100
Contrast	> 10,000:1	> 10,000:1	> 10,000:1

P0.78 is factory-custom and is not a standard Spectrum push SKU. Receiving cards are included in the wall price; sell cards only as spares. Specs from factory COB brief — confirm against the quote before a bid.

SPECTRUM DISPLAY INC

Same team for the wall and the processor.

Azusa warehouse. Spectrum MK Flip-Chip COB and NovaStar on one quote. Selected COB: 3-year warranty plus 5-year local support.

WAREHOUSE

517 N. Virginia Ave.
Azusa, CA 91702 USA

CALL

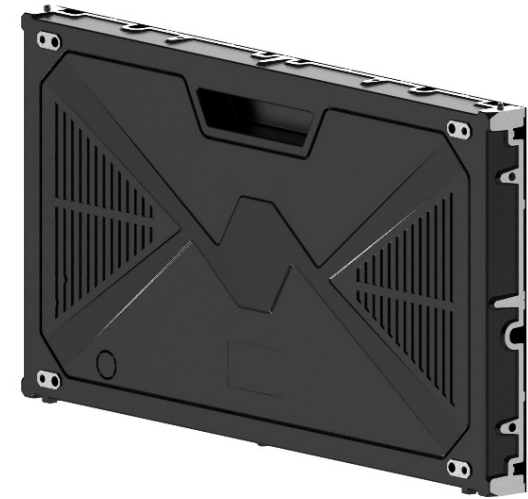
1.844.848.8899
sales@spectrumdisplay.com

WEB

www.spectrumdisplay.com
info@ · support@

COB COVER

3-year warranty
5-year local support

**Ready for NovaStar COEX**

Pair MK cabinets with MX-class processors and CVT fiber converters. Receiving cards ship in the wall. Do not add cards as a line on a new-wall quote.